SIMATS ENGINEERING
CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:40**

Code of Conduct:

I Ganji Madhan Kumar(1924723280certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **G.Madhan Kumar**

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 – Smart Medical Diagnosis System
(a) Propositional Logic Representation
Let,
F = Patient A has Fever
C = Patient A has Cough
R = Patient A has Rash
B = Patient A has Breathlessness
Flu = Patient A has Possible Flu
M = Patient A has Possible Measles
P = Patient A has Possible Pneumonia
Rules:

1. If Fever and Cough then Flu

$F \land C = Flu$

2. If Fever and Rash then Measles

$F \land R = M$

3. If Cough and Breathlessness then Pneumonia

$C \land B = P$

Given facts:
F = True
C = True
B = True

(b) Forward Chaining
In forward chaining, we start with the given facts and apply the rules one by one.
Initial facts:
Fever = True
Cough = True
Breathlessness = True
Step 1:
Rule 1:
$F \land C \rightarrow= Flu$
Here Fever and Cough are both true.
So,
Flu = True
Therefore, Patient A has Possible Flu.
Step 2:
Rule 2:
$F \land R = M$
Fever is true but Rash is not given.

So this rule cannot be applied.

Therefore, Measles cannot be derived.

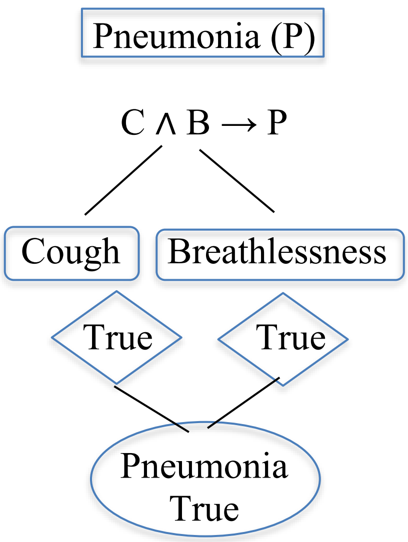
Step 3:
Rule 3:
$C \wedge B = P$
Cough and Breathlessness are both true.

$P = \text{True}$
Therefore, Patient A has Possible Pneumonia.
Final results:
Possible Flu = True
Possible Pneumonia = True
Possible Measles = Not derived

So the diagnoses obtained are **Flu and Pneumonia**.

(c) Backward Chaining
Goal:
Patient A has Pneumonia.
So,
$P = ?$
We check which rule gives Pneumonia.
Rule 3 is:
$C \wedge B = P$
To prove P, we need to prove C and B.
C = Cough → Given fact → True
B = Breathlessness → Given fact → True
Both conditions are true.
Therefore,
$C \wedge B = \text{True}$
Hence,
$P = \text{True}$
So Patient A has Possible Pneumonia.

Goal Reduction Tree



Case Study 2 – Autonomous Traffic Management Agent

(a) Unification

Rule 1:

Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

Given facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Compare the rule with the facts:

x = Ambulance

Therefore,

θ = {x / Ambulance}

So,

Vehicle(Ambulance) ∧ EmergencyType(Ambulance)

Hence:

ClearPath(Ambulance)

(b) Resolution

Goal:

Proceed(CarA)

CNF of Rule 1:

¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

CNF of Rule 2:

¬ClearPath(x) ∨ GreenSignal(x)

CNF of Rule 3:

¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Given:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA,Ambulance)

Step 1:

Vehicle(Ambulance) + EmergencyType(Ambulance)

→ ClearPath(Ambulance)

Step 2:

ClearPath(Ambulance)

→ GreenSignal(Ambulance)

Step 3:

Vehicle(CarA) + Behind(CarA,Ambulance) + GreenSignal(Ambulance)

→ Proceed(CarA)

Therefore:

Proceed(CarA) is true.

(c) Two Emergency Vehicles

The rules can identify two emergency vehicles because Rule 1 uses ∀x.

For example:

Vehicle(Ambulance1)

EmergencyType(Ambulance1)

Vehicle(Ambulance2)
EmergencyType(Ambulance2)
Then:
ClearPath(Ambulance1)
ClearPath(Ambulance2)
But there is no rule to decide which ambulance should get priority if both need the same road.
So we need an extra priority rule.
Example:
HigherPriority(x,y) → GivePriority(x)
And a fact like:
HigherPriority(Ambulance1,Ambulance2)

Case Study 3 – Agricultural Expert Reasoning System
(a) CNF Conversion
Given:
P1: SoilDry → IrrigationNeeded
Remove →:
¬SoilDry ∨ IrrigationNeeded
So,
P1 = ¬SoilDry ∨ IrrigationNeeded
P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
Remove →:
¬(IrrigationNeeded ∧ CropWheat) ∨ ApplyDripMethod
Using De Morgan's law:
P2 = ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
Remove →:
¬(¬ApplyDripMethod ∧ CropWheat) ∨ CropAtRisk
Therefore:
P3 = ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
Facts:
SoilDry = True
CropWheat = True
(b) Resolution Refutation
Goal:
ApplyDripMethod
Negate the goal:
¬ApplyDripMethod
Now:

1. ¬SoilDry ∨ IrrigationNeeded
2. ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
3. SoilDry

4. CropWheat
5. ¬ApplyDripMethod

From 1 and SoilDry:
IrrigationNeeded
From 2, IrrigationNeeded and CropWheat:
ApplyDripMethod
But we already have:
¬ApplyDripMethod
Therefore:
ApplyDripMethod + ¬ApplyDripMethod
= □
So,
ApplyDripMethod is proved.
Simple proof:

SoilDry
↓
IrrigationNeeded
↓
CropWheat + IrrigationNeeded
↓
ApplyDripMethod
↓
¬ApplyDripMethod

(c) If SoilDry is removed
Now only:
CropWheat = True
is given.
Without SoilDry:
IrrigationNeeded cannot be derived.
So ApplyDripMethod also cannot be derived.
For CropAtRisk, we need:
¬ApplyDripMethod ∧ CropWheat
But ¬ApplyDripMethod is not given.
Therefore:
CropAtRisk cannot be concluded.
Final:
IrrigationNeeded → Not derived
ApplyDripMethod → Not derived
CropAtRisk → Not derived

Case Study 4 – Wumpus World Knowledge Agent

(a) Belief State

Given:

Stench at [1,2]

Breeze at [1,1]

Glitter at [2,2]

Using the rules:

Stench[1,2] → Wumpus is adjacent to [1,2]

So,

Wumpus is adjacent to [1,2].

Breeze[1,1] → Pit is adjacent to [1,1]

So,

Pit is adjacent to [1,1].

Glitter[2,2] → Gold[2,2]

So,

Gold is at [2,2].

Derived facts:

- Wumpus is near [1,2]
- Pit is near [1,1]
- Gold is at [2,2]

(b) Forward Chaining

Initial facts:

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Step 1:

Stench[1,2]

→ Wumpus is adjacent to [1,2]

Possible Wumpus locations:

[1,1], [1,3], [2,2]

Step 2:

Breeze[1,1]

→ Pit is adjacent to [1,1]

Possible Pit locations:

[1,2], [2,1]

Step 3:

Glitter[2,2]

→ Gold[2,2]

Therefore:

Gold is at [2,2].

(c) Checking the Path

Path:

[1,1] → [1,2] → [2,2]

From the Breeze at [1,1]:

Pit can be at [1,2].

From the Stench at [1,2]:
Wumpus can be at [2,2].
So:
[1,2] may contain a Pit.
[2,2] may contain a Wumpus.
Therefore, the agent cannot say that the path is safe.
So the path:
[1,1] → [1,2] → [2,2] is not guaranteed to be safe.
The agent should avoid this path until it gets more information.